

O0852 Program Complete Analysis

CNC Macro Program Training Material

Ring-shaped Part Auto Continuous Machining System | FANUC
Macro

Macro Automation — Set material/dimension/conditions by variable
input only

Continuous Machining — Face→Step→Chamfer→Cut cycle auto
repeat

Auto Link — Auto material pull when material runs short

Safety Validation — 30+ alarms to prevent input errors

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1. Overview

O0852 is a FANUC macro program that automatically and continuously machines pipe material on a CNC lathe to mass-produce ring-shaped parts (bushings/bearings).

Current Settings Summary

Item	Material	Finished Part	Production
Specs	CN, OD70 × ID56 Length 543mm	OD64.9 × ID58.3 Length 7.823mm	9.843mm per piece ~6 per pull cycle

Key Features

Macro Automation: Set material, dimensions, cutting conditions by variable input. Multi-product support without code changes

Continuous Machining: Face → Step cut → Chamfer → Cut-off cycle auto repeats

Auto Link: When material runs short, auto-pulls material and continues machining

Safety Validation: 30+ alarm checks prevent input errors, dimension conflicts, safety issues

2. Program Structure

4 sub-programs are called sequentially.

Program	Name	Role	Call Method
O0852	Main Setup	Parameter input, validation, machine config	Direct run
O9001	Main Logic	Length/count calc, alarm check, start machining	M98 P9001
O9002	Main Machining	Face→Step→Chamfer→Cut repeat, Auto Link	M98 P9002
O9003	Remaining	Remaining material pull, re-setup, final cycle	M98 P9003

Call Flow

O0852 (Setup) → O9001 (Calc/Validate) → O9002 (Machining Cycle) → O9003 (Remaining)

Tools Used

Tool	Purpose	Main Process	Note
T01	Boring bar (ID machining)	Rough, Finish, Chamfer	Main tool
T02	Cut-off insert	Facing, Cut-off	Face + Parting
T03	Auto Link	Material pull	Clamp/Unclamp

3. Input Parameters

Operator modification area (Line 4~29, modify #101~#123 only)

3-1. Basic Settings (#101 ~ #108)

Variable	Description	Value	Input Rule
#101	Material length ones digit	43	Range 0~99
#102	Material length hundreds	500	Only 0,100,200,300,400,500
#103	Chuck length (grip)	75	mm unit
#104	Initial face cut	1	Max 40
#105	Material type	1	1=CN, 2=RS, 3=CM
#106	Process type	3	3=Chamfer, 4=Chamfer(ext)
#107	Rough ON/OFF	0	0=OFF, 1=ON
#108	Single mode	1	1=One at a time

Material length: #101 + #102 = Total → Ex: 500 + 43 = 543mm

3-2. Dimension Data (#109 ~ #118)

Variable	Description	Value	Unit
#109	Raw OD (Outside Diameter)	70	mm
#110	Raw ID (Inside Diameter)	56	mm

Variable	Description	Value	Unit
#111	Finished OD	64.90	mm
#112	Finished ID	58.30	mm
#113	Finished Length	7.823	mm
#114	Tip Width (Cut-off)	2.02	mm
#115	ID Radius	0.34	mm
#116	OD Radius	0.36	mm
#117	ID Chamfer	0.37	mm
#118	OD Chamfer	0.60	mm
#119	Length>40mm flag	0	Set 1 if >40mm

3-3. Cutting Conditions (#120 ~ #123)

Variable	Description	Value	Note
#120	T01 RPM	1700	Boring bar speed
#121	T02 RPM	1700	Cut-off insert speed
#122	Pull Distance	1	Auto Link pull distance
#123	Margin Value	0.2	Range 0~3

4. Feed Rate by Material Type

Feed coefficients are auto-determined by material type (#105).

Material	#124 (T01 Rough)	#125 (T01 Step)	#126 (T01 Chamfer)	#127 (T02 Cut)
CN (#105=1)	0.15	0.12	0.15	0.09
RS (#105=2)	0.13	0.08	0.05	0.08

Material	#124 (T01 Rough)	#125 (T01 Step)	#126 (T01 Chamfer)	#127 (T02 Cut)
CM (#105=3)	0.13	0.09	0.09	0.09
Default	0.15	0.10	0.10	0.10

Feed Rate Formula

F value = RPM × Feed coefficient

Calculation Examples (RPM = 1700)

Process	Material	Calculation	Result
T01 Rough	CN	1700×0.15	F255 mm/min
T01 Step	CN	1700×0.12	F204 mm/min
T02 Cut-off	CN	1700×0.09	F153 mm/min
T01 Rough	RS	1700×0.13	F221 mm/min

RS material has lower feed rates overall (higher hardness requires conservative cutting).

5. Advanced Settings & Safety

DANGER: STOP EDITING — This section is auto-configured. DO NOT modify!

5-1. System Variables (#500 series)

Variable	Description	Value	Role
#530	Safety length	12	Safety distance from chuck end
#531	Min remaining length	15	Min 15mm must stay in jaw after machining → Short remaining = workpiece ejection risk
#532	Remaining-safety offset	1.03	Calibration value for accurate positioning during last Auto Link pull
#506	Z rough max depth	55	Max Z-direction length per pass
#507	Z rough clearance	1	Z-direction clearance
#508	T01 nose R	0.2	Tool tip radius
#509	Rough interval	0.4	X-direction interval
#510	Chamfer interval	1	Chamfer cut interval

5-2. RS40 / Small Diameter Safety

**RS40 material (#105=2) or OD < 30mm (#109<30) → #129=15
auto-set**

**Adds 15mm residual offset to prevent workpiece ejection due to
low rigidity.**

6. Machining Cycle Detail

6-1. Overall Flow

Step	Process	Tool	Main Action
(1)	Coordinate set	-	G10 L2 P0 Z[#103] — Set G54 work Z origin
(2)	Facing	T02	Face end surface (OD→ID direction)
(3)	Step cut	T01	Rough(option) + Finish — Z-direction cut
(4)	Chamfer	T01	OD-R, ID-R chamfer
(5)	Cut-off	T02	Part separation + M12 count
	Repeat		Repeat (3)~(5) for count (#517)

Key Calculation Variables

Formula	Meaning	Current Value
$\#505 = \#113 + \#114$	Unit per piece (length + cut width)	9.843mm
$\#500 = \#103 - \#530$	Effective machining length	63mm
$\#517 = \text{FIX}[\#500/\#505]$	Pieces per pull cycle	6 pcs
$\#518 = \#517 \times \#505$	Total machining length per cycle	59.058mm
$\#140 = \#101 + \#102 - \#129$	Total material length (with offset)	543mm

7. Auto Link & Remaining Material

7-1. Auto Link Sequence (N210)

Core function that automatically pulls material when it runs short.

Step	Code	Action	Description
(1)	M05	Spindle stop	Stop rotation for safety
(2)	G00 T03, X0	Tool select	Auto Link tool, center move
(3)	Z[-#522+#122]	Approach pull pos	Move to pull position at F200
(4)	M[#132]	Auto Loader CLOSE	Loader grips material
(5)	M69 + G04 P2500	Chuck UNCLAMP	Chuck releases (2.5s wait)
(6)	W[length+marg in]	Pull material	Pull material forward
(7)	M68 + G04 P2500	Chuck CLAMP	Chuck re-grips (2.5s wait)
(8)	M[#131]	Auto Loader OPEN	Loader releases
(9)	G00 Z100→GO TO100	Return	Safe pos → restart from facing

SAFETY WARNING: Chuck opens and closes during Auto Link. NEVER touch material or tools during this sequence! Do NOT reduce G04 wait times.

8. Alarm Code Table

When an error occurs, check the alarm number (#3000) and refer to the table below.

8-1. Machining Logic Alarms

Alarm	Condition	Meaning	Action
1	#140 > 580	Material length > 580mm	Check #101, #102
2	#109 < #111	Raw OD < Finished OD	Check #109, #111
3	#110 > #112	Raw ID > Finished ID	Check #110, #112
4	#125 >= 0.14	T01 step feed too high	Check material type (#105)
5	#102 unit error	Not in 100s unit	Use 0/100/200/300/400/500
6	#101 > 99	Ones digit out of range	Change to 0~99
7	#140 <= #103	Material too short	Check material length
10	#104 > 40	Init face cut too large	Set #104 to 40 or less

8-2. Safety Validation Alarms

Alarm	Condition	Meaning	Action
203	OD <= ID	Raw OD <= Raw ID	Check #109, #110
204	FIN OD <= FIN ID	Finished OD <= ID	Check #111, #112

Alarm	Condition	Meaning	Action
205	#109 <= 0	OD is 0 or negative	Check #109
206	#120 <= 0	T01 RPM is 0 or less	Check #120
207	#121 <= 0	T02 RPM is 0 or less	Check #121
208	#114 >= #113	Tip width >= Fin length	Check #114, #113

8-3. Missing Variable Alarms (101~130, 506~532)

Alarm 101~130 or 506~532: The corresponding variable (#number) is not entered.

Ex) Alarm 109 → #109 (Raw OD) not entered → Enter the value.

9. Machine M-Code Mapping

M-codes for Auto Loader/Boring bar are auto-mapped by machine number (#130).

Mach ne No.	Name	AL OPEN (#131)	AL CLOSE (#132)	BR UP (#133)	BR DOWN (#134)
1	AL-1	M56	M55	M54	M53
4	HA-4	M52	M51	M54	M53
5	AL-5	M171	M170	M53	M54
7	AL-7	M64	M63	M53	M54
8	AL-8	M64	M63	M54	M53
9	AL-9	M63	M64	M53	M54
10	AL-10	M64	M63	M54	M53
13	HA-S3	M56	M55	M54	M53
14	HA-S4	M56	M55	M54	M53

AL = Auto Loader | BR = Boring bar

To change machine: only modify #130. M-codes auto-map.

**BR UP/DOWN differs by machine. AL-5, AL-7, AL-9 are reversed.
Always modify #130 only. Never edit M-codes directly.**

10. Operator Precautions

10-1. Required Checks

No.	Check Item	Detail
1	Editable area	#101~#123 only. NEVER edit after DANGER section
2	#119 check (long parts)	If #113 > 40mm, must set #119=1
3	#102 input format	Only 0,100,200,300,400,500 Other values → Alarm 5
4	#101 input range	0~99 only 100+ → Alarm 6
5	Machine change	Only modify #130 Never edit M-codes directly

10-2. Safety Rules

- (1) NO intervention during Auto Link — NEVER touch material/tools during chuck open/close!
- (2) RS40 material — 15mm safety offset auto-applied, do NOT disable
- (3) Small OD (less than 30mm) — Same safety offset applied
- (4) On alarm — Check alarm number → Refer to code table → Fix variable → Restart
- (5) G04 wait times — Do NOT reduce Auto Link wait times

10-3. Pre-Machining Checklist

	Check Item
☐	Material type (#105) matches actual material?
☐	Raw OD/ID (#109, #110) matches measured values?
☐	Finished dimensions (#111~#118) match drawing?

☐	Machine number (#130) matches actual machine?
☐	RPM (#120, #121) suitable for material/tool?
☐	First run in single mode (#108=1) for test?
☐	Auto Loader operation verified?
☐	Cut-off insert (T02) wear checked?

O0852 Program Key Summary

1. Operators only modify variables #101~#123 (material, dimensions, conditions)
2. Feed rates auto-set by material type (CN/RS/CM)
3. Face → Step → Chamfer → Cut-off cycle auto-repeats
4. Auto Link pulls material when it runs short
5. 30+ alarms prevent input errors and safety issues
6. 9 machine M-codes auto-mapped (change #130 only)

If you have questions, please ask anytime